

Project Review of Project Milestone 3 Skin cancer and Sunscreen by Rituraj Kumar

I've reviewed the PowerPoint presentation prepared by Rituraj Kumar on "**The Role of Sunscreen in Skin Cancer Prevention**" as part of the project milestone three and prepared this document with my findings. My review addresses the following five questions:

1. Is the theory or question clear and understood?

The project addresses a clear research question: "Do sunscreens protect from skin cancer?" The slides explore how sunscreen interacts with various types of skin cancers like Basal Cell Carcinoma (BCC), Squamous Cell Carcinoma (SCC), and Melanoma, and consider the influence of factors like skin color, UV exposure, and SPF usage. The framing of the question is understandable and sets the stage for thoughtful analysis.

2. Strengths of the outlined plan:

- Well-defined introduction connecting sunburn with skin cancer types.
- Recognition of various influencing factors (e.g., skin tone, location, altitude).
- Balanced view of sunscreen's effectiveness, acknowledging its limits and potential side effects.
- Use of practical recommendations (e.g., lower SPF used correctly can reduce risks).

3. Areas of opportunity to improve the analysis:

- The presentation could benefit from including more specific scientific data or citations from studies.
- Differentiation between chemical vs. mineral sunscreens would clarify the health impact.
- Analysis could explore how sunscreen affects Vitamin D synthesis across populations.
- Data visualizations (e.g., graphs, charts) would enhance understanding and support key points.

4. Assumptions that should be investigated more:

- The assumption that higher SPF equals higher chemical absorption is plausible but needs quantifiable evidence.
- That sunscreen has little effect on BCC, and that melanoma may not apply universally should be backed by region-specific studies.
- Assumption of skin vulnerability based solely on skin tone should also consider genetic, lifestyle, and environmental factors.

5. Ethical implications to consider:

- It's important to consider equity in access to sun protection across different socioeconomic groups.

- Recommendations should be sensitive to varying levels of sun exposure due to occupation or cultural clothing norms.
- Transparency about the potential risks of chemical absorption should be handled carefully to avoid misinformation or sunscreen avoidance.
- Encouraging informed choices over fear-based messaging is key.